

VILA REAL FLORES, Portugal

WMO: 085660

Lat: 41.317N

Lon: 7.733W

Elev: 481

StdP: 95.68

Time Zone: 0.00 (EUW)

Period: 82-92

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.8	-0.6	-5.2	2.6	1.3	-3.7	2.9	1.6	8.1	9.9	7.1	8.3	0.6	100	0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	33.4	21.8	31.4	20.6	29.7	19.8	22.4	32.3	21.3	30.4	20.2	28.6	2.2	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	14.3	28.7	17.7	13.5	26.5	16.9	12.8	24.9	68.0	32.7	63.7	30.5	59.8	28.7	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.5	5.3	4.5	-3.4	36.9	1.5	1.5	-4.5	37.9	-5.3	38.8	-6.1	39.6	-7.2	40.7		
			-4.2	23.7	1.5	1.1	-5.3	24.5	-6.2	25.1	-7.1	25.8	-8.2	26.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.4	5.3	7.6	9.9	11.1	14.3	18.5	21.6	21.2	19.9	14.1	9.7	7.2
	DBStd	6.39	3.04	2.80	2.82	3.42	3.66	3.47	3.55	3.00	3.38	3.09	3.26	2.91
	HDD10.0	434	146	73	38	27	6	1	0	0	0	3	44	96
	HDD18.3	2133	403	299	263	217	134	40	10	8	21	134	258	346
	CDD10.0	1673	1	7	33	60	138	255	359	346	298	131	36	9
	CDD18.3	331	0	0	0	0	8	44	110	95	68	4	0	0
	CDH23.3	3322	0	0	1	2	94	446	1158	962	619	39	0	0
	CDH26.7	1208	0	0	0	0	13	128	488	363	213	3	0	0
Wind	WSAvg	1.6	1.3	1.7	1.9	1.9	1.7	1.7	1.6	1.7	1.4	1.3	1.3	1.3
Precipitation	PrecAvg	1112	160	170	97	90	70	53	15	16	49	108	125	160
	PrecMax	1812	444	449	357	263	190	238	74	68	162	275	459	619
	PrecMin	615	0	23	0	4	10	3	0	0	2	1	0	9
	PrecStd	294	112	127	88	66	45	53	20	18	40	79	99	143
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	16.6	21.5	23.5	28.5	32.0	36.1	36.1	33.8	26.6	18.3	16.0
		MCWB	10.5	12.1	14.3	14.9	19.4	21.0	23.6	23.2	21.5	18.5	14.6	13.5
	2%	DB	11.8	14.4	19.3	21.5	26.0	30.0	33.8	33.1	31.2	24.2	17.2	14.2
		MCWB	9.9	10.5	13.1	14.1	17.8	19.8	22.2	21.3	19.9	17.0	14.3	11.9
	5%	DB	10.9	12.9	17.2	19.7	23.7	28.0	31.7	30.7	29.2	22.0	15.7	12.7
		MCWB	9.2	10.1	11.6	13.2	16.6	18.6	21.1	20.1	19.0	15.9	13.2	10.8
	10%	DB	9.9	11.9	15.2	17.5	21.3	26.0	29.8	28.7	27.0	19.6	14.4	11.5
		MCWB	8.3	9.5	10.5	12.2	15.3	17.8	20.2	19.3	18.3	14.6	12.5	9.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	12.9	14.7	15.8	19.8	22.1	24.1	23.6	22.2	18.9	15.9	13.9
		MCDB	12.3	14.5	20.8	21.1	27.1	30.9	35.6	35.2	32.7	25.0	17.3	15.1
	2%	WB	10.4	12.0	13.4	14.8	18.1	20.3	22.7	22.0	20.4	17.3	14.5	12.1
		MCDB	11.3	13.3	18.6	20.5	25.4	28.8	32.7	31.9	30.0	23.1	16.3	13.7
	5%	WB	9.6	10.8	12.2	13.7	16.9	19.1	21.6	20.7	19.4	16.2	13.7	11.1
		MCDB	10.5	12.4	16.5	18.4	23.1	27.0	30.7	29.5	27.9	20.9	15.5	12.3
	10%	WB	8.5	9.8	11.0	12.6	15.6	18.1	20.5	19.6	18.5	15.1	12.7	10.1
		MCDB	9.6	11.5	14.3	16.7	20.6	25.2	28.9	27.6	25.9	19.0	14.1	11.2
Mean Daily Temperature Range	5% DB	MDBR	5.9	6.6	8.9	8.5	10.2	11.3	12.9	13.5	12.1	8.8	5.9	5.4
		MCDBR	6.4	8.6	12.3	12.0	14.2	14.7	15.3	16.3	15.5	13.4	6.8	5.5
		MCWBR	4.8	5.5	7.1	6.4	7.6	7.0	7.2	7.4	7.2	7.1	4.3	4.0
	5% WB	MCDBR	5.0	6.8	11.2	11.1	13.3	14.0	14.7	15.5	14.5	11.8	5.6	4.7
		MCWBR	4.4	4.9	6.7	6.0	7.4	7.2	7.3	7.3	7.1	6.6	3.9	3.8
Clear-Sky Solar Irradiance	taub	0.303	0.315	0.347	0.351	0.360	0.362	0.349	0.362	0.350	0.340	0.312	0.300	
	taud	2.538	2.493	2.416	2.409	2.408	2.414	2.432	2.391	2.449	2.498	2.537	2.542	
	Ebn at Noon	860	900	903	920	915	910	918	898	887	857	839	832	
	Edh at Noon	71	86	104	113	116	115	112	113	100	85	71	66	
All-Sky Solar Radiation	RadAvg	1.79	2.84	3.92	5.05	6.26	7.12	7.42	6.62	5.06	3.17	2.03	1.60	
	RadStd	0.29	0.42	0.62	0.63	0.65	0.45	0.32	0.34	0.37	0.41	0.38	0.24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page